

EVIDENCE COMPLETE

1BILL VPN Evidence Pack

Fintap endpoint validation, Palo Alto interoperability test, and production fault-domain proof

Prepared for	1LINK / 1BILL network and security teams
Prepared by	Zynotch PVT Limited - Fintap
Evidence date	17 September 2026 (PKT); server captures include timezone offsets
Production peer	1LINK 103.248.140.4 <-> Fintap 178.238.236.126
Scope	IKEv2 Phase 1 does not establish in production

Executive finding

The Fintap endpoint receives and correctly answers a real Palo Alto IKEv2 initiation using the exact 1BILL Phase 1 proposal. In production, Fintap has received **zero** packets from the documented 1LINK peer IP. The current production failure is therefore before PSK authentication, IKE identities, proxy IDs, or Phase 2.

Control	Observed result	Meaning
Fuel Palo Alto -> Fintap	9 IKE_SA_INIT requests received; 9 responses sent	Fintap listener, route and responder work
Exact IKE proposal	AES-256 / SHA-256 / PRF-SHA256 / group 19 selected	The 1BILL document proposal is accepted
Production 1LINK -> Fintap	0 inbound packets from 103.248.140.4	Production never reaches crypto/auth evaluation
Fintap -> production 1LINK	29 IKE packets sent with no received reply	Peer/path does not answer
Independent control SA	Established with same algorithm suite	StrongSwan and kernel IPsec path are functional

Confidence boundary

95% confidence that the Fintap VPN host is functioning and the production block is on the 1LINK-controlled side or the network/security path observable by 1LINK. The exact 1LINK device or rule cannot be named until 1LINK supplies an Untrust capture and ikemgr log from the active firewall.

PALO ALTO GUI EVIDENCE - ACTUAL FUEL VM CAPTURE

Actual PAN-OS 11.1.4 dashboard evidence

The image below is a direct print export of the live Palo Alto VM-Series GUI used in the Fuel interoperability test. It is not a mock-up and it is not represented as a screenshot from 1LINK's production firewall. The full image is retained so the PAN-OS navigation, device model/version and contemporaneous system-log context remain visible.

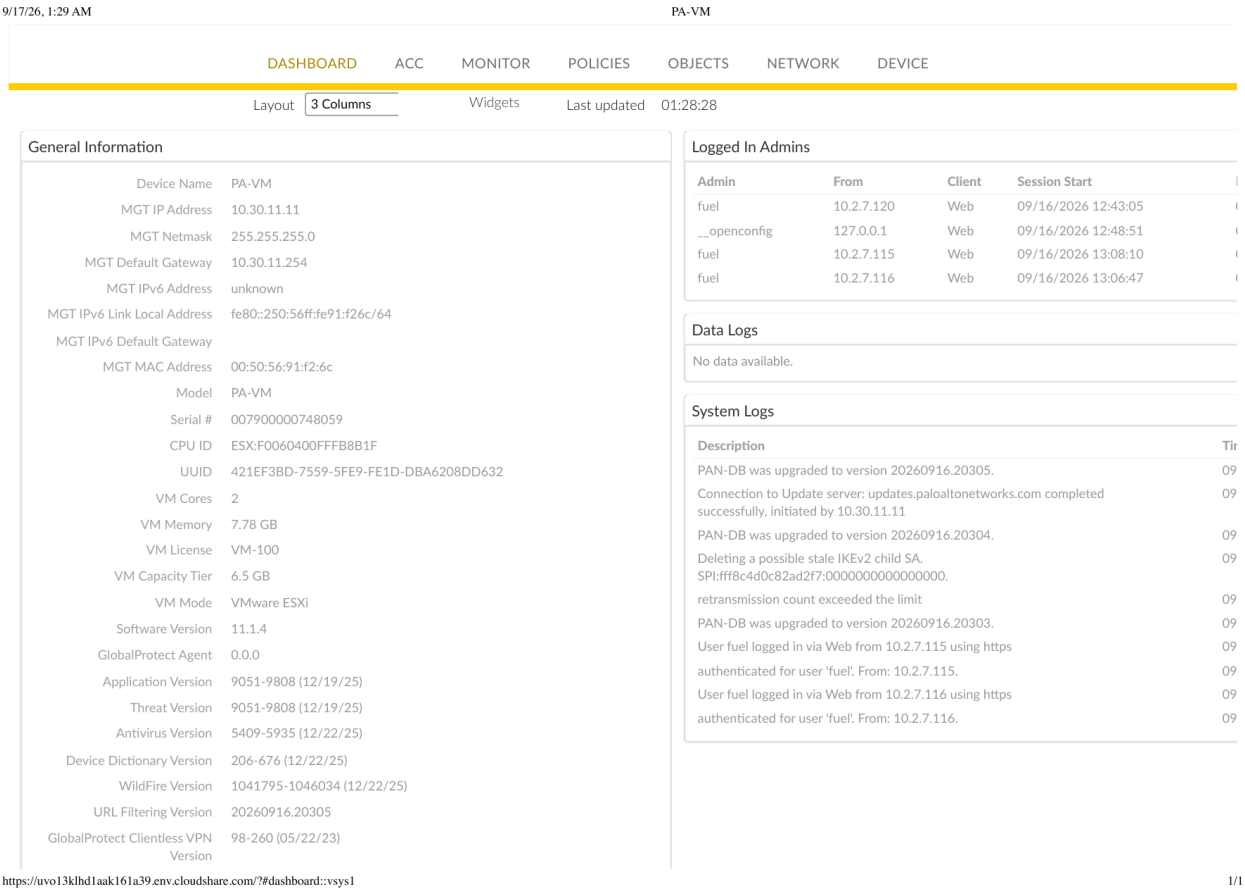


Figure 1. Fuel PAN-OS VM-Series dashboard after the IKEv2 test. The System Logs panel visibly records a stale IKEv2 child SA and 'retransmission count exceeded the limit'.

Visible GUI evidence	Technical meaning
Software Version 11.1.4	The test used a current PAN-OS interface, not an obsolete GUI simulation.
Deleting a possible stale IKEv2 child SA	PAN-OS created IKEv2 state for the test attempt.
retransmission count exceeded the limit	The initiator did not accept/receive the return exchange.

Evidence handling

The GUI capture contains no PSK, private key, authentication secret or IPsec key material. It is supporting lab evidence; the production-side proof still requires 1LINK's own active-firewall Untrust capture and ikemgr log.

EVIDENCE 1 - A REAL PALO ALTO REACHES FINTAP AND FINTAP ANSWERS

Correlated Palo Alto and StrongSwan proof

The same IKE initiator SPI appears in the PAN-OS system log and in Fintap's packet capture. This correlates the two independent observations even though the lab systems use different clock/timezone settings.

PAN-OS field	Live value
Event	ikev2-nego-ike-start
Gateway	FUEL-FINTAP-GW
Role	Initiator, non-rekey
Flow	172.16.1.1:500 -> 178.238.236.126:500
Initiator SPI	fff8c4d0c82ad2f7
PAN status after test	IKE Info: 0 items; Phase 1 not established

root@178.238.236.126 - StrongSwan journal + tcpdump

SANITIZED LIVE CAPTURE

```
2026-09-16T22:03:51+02:00 received packet:
  from 38.104.95.242[10400] to 178.238.236.126[500] (240 bytes)
parsed IKE_SA_INIT request 0 [ SA KE No N(NATD_S_IP) N(NATD_D_IP) ]
38.104.95.242 is initiating an IKE_SA
selected proposal:
  IKE:AES_CBC_256/HMAC_SHA2_256_128/PRF_HMAC_SHA2_256/ECV_256
remote host is behind NAT
generating IKE_SA_INIT response 0 [ SA KE No ... ]
sending packet:
  from 178.238.236.126[500] to 38.104.95.242[10400] (256 bytes)
tcpdump initiator cookie: fff8c4d0c82ad2f7 -> 0000000000000000
```

What this proves

Fintap accepted the exact 1BILL IKE proposal and generated a valid IKE_SA_INIT response. The PSK and traffic selectors are not used until later exchanges, so this is direct proof of Phase 1 message-1 interoperability.

EVIDENCE 2 - WHY THE FUEL LAB DID NOT COMPLETE THE FULL TUNNEL

The return packet was lost after leaving Fintap

CloudShare translated the Palo Alto source to a shared public address and a changing high UDP port. Fintap answered every observed request to the translated source port, but PAN-OS never progressed beyond IKE_SA_INIT and retransmitted the identical initiator SPI.

tcpdump - public IKE traffic

SANITIZED LIVE CAPTURE

```
22:03:51 38.104.95.242:10400 -> 178.238.236.126:500 IKE_SA_INIT request
22:03:51 178.238.236.126:500 -> 38.104.95.242:10400 IKE_SA_INIT response
22:03:56 38.104.95.242:10400 -> 178.238.236.126:500 retransmit
22:03:56 178.238.236.126:500 -> 38.104.95.242:10400 response retransmit
22:04:06 38.104.95.242:10400 -> 178.238.236.126:500 retransmit
22:04:26 38.104.95.242:10400 -> 178.238.236.126:500 new attempt
22:05:06 38.104.95.242:10401 -> 178.238.236.126:500 new NAT mapping
22:06:06 38.104.95.242:10402 -> 178.238.236.126:500 new NAT mapping
CAPTURE TOTAL: 12 packets, 0 dropped by the Fintap kernel
```

Stage	Observed	Assessment
Palo creates IKE_SA_INIT	Yes	PAN-OS gateway and proposal are active
CloudShare sends it publicly	Yes	Request reaches Fintap
Fintap accepts proposal	Yes	Exact crypto suite selected
Fintap sends response	Yes	Response visible leaving eth0
Palo processes response	No	CloudShare/NAT return path is the lab limitation
IKE_AUTH / PSK	Not reached	Cannot be blamed for this lab stop

Important limitation

The Fuel environment proves Palo-to-Fintap message-1 compatibility and Fintap's response. It does not prove the lab's shared Internet NAT can carry a complete site-to-site VPN. This limitation does not weaken the production finding, where no 1LINK request reaches Fintap at all.

EVIDENCE 3 - PRODUCTION 1LINK TRAFFIC NEVER REACHES FINTAP

Production evidence: zero inbound packets from the documented peer

The audit below queried the StrongSwan service journal from 1 September 2026 onward. It counts packets, not configuration entries. Loading a PSK or connection does not increase the inbound count.

**StrongSwan production packet-count audit**

SANITIZED LIVE CAPTURE

```
root@178.238.236.126:~# production-ike-audit
Query window: 2026-09-01 through 2026-09-17
Expected 1LINK public peer: 103.248.140.4
```

```
INBOUND_FROM_1BILL=0
```

```
OUTBOUND_TO_1BILL=29
```

```
Control window - Fuel Palo test:
```

```
FUEL_INBOUND=9
```

```
FUEL_RESPONSES=9
```

**StrongSwan production attempt - 14 Sep 2026**

SANITIZED LIVE CAPTURE

```
2026-09-14T15:02:10+02:00 initiating IKE_SA onebill[318] to 103.248.140.4
15:02:10 sending 178.238.236.126[500] -> 103.248.140.4[500] (272 bytes)
15:02:14 sending 178.238.236.126[500] -> 103.248.140.4[500] (272 bytes)
15:02:21 sending 178.238.236.126[500] -> 103.248.140.4[500] (272 bytes)
15:02:34 sending 178.238.236.126[500] -> 103.248.140.4[500] (272 bytes)
15:02:57 sending 178.238.236.126[500] -> 103.248.140.4[500] (272 bytes)
No corresponding 'received packet: from 103.248.140.4' record exists.
```

Protocol conclusion

With zero inbound IKE_SA_INIT packets, PSK comparison, IKE identities, proxy IDs, PFS, Phase 2 and HTTPS are not reached. A mismatch in those later parameters cannot produce the observed zero-packet condition.

EVIDENCE 4 - FINTAP SERVICE READINESS AND INDEPENDENT CONTROL

The endpoint is listening and an exact-profile control SA is established



Fintap IKE listeners

SANITIZED LIVE CAPTURE

```
root@178.238.236.126:~# ss -ltnp | grep -E ':(500|4500) '
UNCONN 0 0 0.0.0.0:500 0.0.0.0:* users:((charon-systemd,pid=63160))
UNCONN 0 0 0.0.0.0:4500 0.0.0.0:* users:((charon-systemd,pid=63160))
UNCONN 0 0 [::]:500 [::]:* users:((charon-systemd,pid=63160))
UNCONN 0 0 [::]:4500 [::]:* users:((charon-systemd,pid=63160))
```



Independent exact-crypto control SA

SANITIZED LIVE CAPTURE

```
azure-test: #399, ESTABLISHED, IKEv2
local 178.238.236.126[4500]
remote 20.187.97.214[4500]
AES_CBC-256/HMAC_SHA2_256_128/PRF_HMAC_SHA2_256/ECP_256
azure-test-https: #26, INSTALLED, TUNNEL-in-UDP
ESP:AES_CBC-256/HMAC_SHA2_256_128/ECP_256
local 172.31.254.10/32
remote 10.1.1.4/32
```

Parameter	1BILL documented value	Fintap / tested value	Result
IKE version	IKEv2 only	IKEv2	Match
IKE encryption	AES-256-CBC	AES_CBC_256	Match
IKE integrity	SHA-256	HMAC_SHA2_256_128	Match
IKE PRF	PRF 256	PRF_HMAC_SHA2_256	Match
DH group	Group 19	ECP_256 / group 19	Match
IKE lifetime	28,800 seconds	28,800 seconds	Match
ESP	AES-256 / SHA-256	AES_CBC_256 / SHA2_256	Match
PFS	Group 19	ECP_256 / group 19	Match

The control tunnel remained established while the temporary Fuel test configuration was removed. This independently confirms that the StrongSwan process, UDP listeners, algorithm implementation, NAT-T handling and kernel IPsec stack are operational.

TECHNICAL INTERPRETATION

Where the production exchange stops

Protocol step	Expected direction	Production status	Settings evaluated here
1. IKE_SA_INIT request	1LINK -> Fintap	NOT OBSERVED	IKE version, proposal, DH, route/ACL
2. IKE_SA_INIT response	Fintap -> 1LINK	Cannot occur	Chosen IKE proposal
3. IKE_AUTH	Both directions	Not reached	PSK and IKE identities
4. Child SA	Both directions	Not reached	Proxy IDs, ESP, PFS
5. Application	Protected traffic	Not reached	TCP 443, SNI, allowlist

What can still be wrong

- 1LINK has not initiated even though it owns initiation for this one-ended design.
- Passive Mode is enabled on the 1LINK Palo Alto gateway, so it only responds and never initiates.
- The customer gateway is missing, disabled, uncommitted, pushed to the wrong template, or configured on the passive HA node.
- The active peer IP, local interface, route, public NAT address, whitelist or upstream ACL is incorrect.
- UDP 500/4500 is blocked before the request reaches Fintap or before a response returns.
- The running gateway uses a different IKE version than the supplied IKEv2 document.

What the evidence rules out as the current blocker

Suspected item	Why it is not the current blocker
PSK	Used in IKE_AUTH, after the missing initial request/response
IKE ID	Authenticated after IKE_SA_INIT
Proxy ID / selectors	Evaluated for the Child SA, later than Phase 1 message 1
Phase 2/PFS	Evaluated only after an IKE SA exists
HTTPS / TCP 443	Application layer after IPsec is installed
Old vs new Palo GUI	PAN-OS 11.1.4 accepted the exact documented proposal

Calibrated conclusion

This evidence does not claim which exact 1LINK appliance or rule is faulty. It proves the fault boundary: no production IKE request from 103.248.140.4 reaches Fintap, while a real Palo Alto request does reach Fintap and is answered.

REQUIRED ACTION FROM 1LINK

One synchronized test will identify the exact failing device

Agree on an exact timestamp and timezone. Start the Fintap capture first. Then 1LINK must initiate from the active Palo Alto firewall and preserve the following outputs.

**Commands requested from the active 1LINK Palo Alto**

SAFE TO SHARE - DOES NOT REVEAL THE PSK

```
show vpn ike-sa gateway <gateway-name>
show vpn ipsec-sa tunnel <tunnel-name>
less mp-log ikemgr.log
test vpn ike-sa gateway <gateway-name>
test vpn ipsec-sa tunnel <tunnel-name>
```

Evidence 1LINK must provide	Acceptance condition
Untrust packet capture	Shows 103.248.140.4 -> 178.238.236.126 on UDP 500/4500
Return traffic in same capture	Shows whether Fintap's response arrives back at the Palo
System log event	ikev2-nego-ike-start on the active gateway
ikemgr log excerpt	Includes the exact attempt timestamp and failure reason
Running gateway screen	IKEv2 only; Passive Mode off if 1LINK initiates; correct interface/peer
HA and commit status	Output is from the active firewall and the candidate change is committed
Panorama / PAN-OS versions	Confirms no version-specific interpretation issue

Meeting statement

Please demonstrate an IKE_SA_INIT packet leaving your active Palo Alto toward 178.238.236.126 and provide the matching Untrust capture. Our server recorded zero inbound packets from 103.248.140.4, while it immediately received and answered every IKE_SA_INIT generated by a real Palo Alto VM.

Evidence handling and cleanup

- No PSK, private key, API credential, database credential or IPsec key material is included in this pack.
- The temporary Fintap Fuel connection and combined-load files were removed after testing.
- The original onebill, laptop-test and azure-test connections were restored and verified loaded.
- The Fuel lab is ephemeral and its Palo Alto configuration disappears when the lab expires.
- Raw log timestamps retain their source timezone. The shared SPI is the correlation key across systems.

Prepared from live observations on the Fuel Palo Alto VM-Series firewall and the production Fintap StrongSwan host. This pack is a sanitized technical evidence summary, not a substitute for 1LINK's own active-firewall capture.